1.09 Transferring Fluids

1. **Purpose**

   The purpose of the procedure is to identify the hazards and define the steps in transferring drilling fluids from a Vacuum Truck to a drilling rig.

2. **Scope**

   This procedure is intended for use for all Silverpoint employees.

3. **Hazard Assessment & Identification**

   - Pressurized Liquids
   - Pressurized Air
   - Flying Debris
   - Spills
   - Pinch Points
   - Slips, Trips, Falls
   - Excessive Noise
   - Equipment Damage
   - Injury Co-Workers

4. **EHS Requirements**

   - Fire retardant Coveralls.
   - Steel Toed Boots.
   - Safety Glasses.
   - Rubber Gloves.
   - Hearing Protection
   - Hardhat

5. **References**

   - Silverpoint Corporate Codes of Practice
   - Alberta OH&S

- Silverpoint Task List
- Silverpoint THA 1.09

6. **Procedure**

It is critical that the vacuum truck operator stays at the controls of the vacuum truck during the transfer procedure. Do not allow anything or anyone to distract you from the job at hand. It is very possible to hurt someone or create a very large mess if you do not follow the procedure and take your time.

6.1. Indentify the tank that the rig wants the fluid transferred into. The rig may want to start the next hole with some of the mud from the last hole. There are many reasons for this, and it really doesn't matter why, you just need to do it right or you can have a very big mess on your hands and a very upset rig crew.

6.2. Safely back into transfer area, use a spotter if necessary. **(SOP 2.01 Backing)**

Refer to Connecting and Disconnecting Hoses SOP 1.02 for sequence of connecting or dis-connecting hoses.

6.3. Choose a good quality hose and connect it to the vacuum truck 6 inch spread valve, so you can get all the fluid off the truck.

ALWAYS connect your HOSES starting at the Vac Unit and ending at the Tank, sump or other.

6.4. Connect other end to rig tank cam lock.

6.4.1. If a cam lock is not available the hose end must be properly secured using a tie down or rope, don't use bungee cord to secure the hose end, they can stretch and break causing the hose end to swing around.

6.4.2. If possible, gravity feed the fluid into the cellar and use the trash pump to transfer the fluid over the shaker.

6.4.3. Whip Checks are required where Vac Tank pressure will be or is used to unload Product.

6.5. Before starting the transfer procedure, you must have a good idea of how much fluid you have on the truck, you will need to compare that to the rig tank level to avoid spraying the rig & workers with mud.

6.5.1. Gravity feed the fluid until you need pressure to move the remaining fluid.

6.5.2. Use extreme caution when unloading fluid from truck, you will need to use pressure to unload the fluid and when the tank is empty you are left with a tank full of pressure that will want to go somewhere.

6.5.3. Always use a good hose with no cracks so you don't have any leaks, make sure cam locks are completely closed, secure the end of the hose as good as you possibly can, use a cam lock connection if there is one available.

6.6. Open the valve on the rig if applicable then open the 6-inch valve on the vacuum truck.

6.7. Place 4-way valve in pressure and cycle the hibon intermittently to create enough pressure to move the fluid.

6.7.1. It is very important that you use as little pressure as possible for transferring fluid because of the amount of air that builds up inside the tank.

6.7.2. Be sure you are standing near the valve control so you can close it as soon as the fluid is off; be sure to have some one watch the end of the hose if you can't see it yourself.

6.8. Monitor and maintain enough pressure to keep fluid flowing at a slow, steady rate. Do Not Rush, in order to complete this procedure safely you must go slowly and take your time, if you rush, you could have a spill.

6.9. Watch the vacuum hose for any signs of movement, as the tank empties the air in the tank will start to mix with the mud that is coming out of the tank, and this will cause the vacuum hose to start jumping. **Tie off your hose.**

6.9.1. It is not necessary to completely empty the tank; in fact, if you can keep a small amount of mud on the truck you are less likely to have a large air and mud release from the end of the vacuum hose.

6.10. Immediately close the 6-inch spread valve and release the pressure from the vacuum tank.

6.11. Close rig tank valve if applicable. Put vacuum truck on vacuum and open the 6 inch, slowly remove cam lock from rig to clear the vacuum hose.

6.12. Once the hose is clear, close 6-inch valve and remove the hose from the vacuum truck.

Remember: if you are in doubt about what to do, call your supervisor